Using Virtual Reality to TeachMedical Skills:
A Literature review

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Abstract

The study explores virtual reality (VR) in medical training and its contribution to surgical accuracy, diagnostics, and patient care. Fourth-year medical students underwent a 360-degree simulation of traditional high-fidelity cases, and surgical accuracy and diagnostics competency both increased with an improvement in performance at a significant level. The scalability and cost-effectiveness of VR, along with safer and easier training and its incorporation into the curriculum for overall skill development, have been stressed in the study. It concludes that virtual reality is a breakthrough in medical training, providing future professionals with a skillset for enhancing patient care. There is a suggestion for future studies and investments in virtual reality technology, including Augmented Reality integration and long-term evaluation of learning outcomes.

Introduction:

Overview: VR is genuinely revolutionizing medical practitioner training by providing a safe environment for constantly practicing complex procedures in a simulator. This provides medical students with hands-on experience without risks, closing the gap between their theoretical learning and clinical application.

Objective: This elaborates on the research, which attempts to assess the effect of VR on developing surgical skills, diagnostic capabilities, and patient care management related to medical education programs.

Significance: The findings from this study can also contribute to future curricula that involve VR in medical training.

Methodology

Conversion to VR: Converted conventionally, high-fidelity simulation problem cases into 360-degree VR technology, fully immersing the students in real-life scenarios when placed at decision-critical points modeling real clinical settings (Chinnock, 1994).

Participants: This study involved interactions by fourth-year students of medics with the VR simulations. Feedback responses regarding the training performance were collected through surveys to determine whether the training was effective.

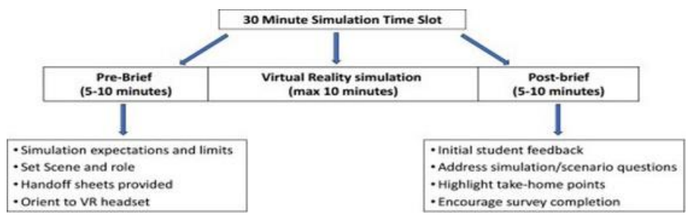

Figure 1: Virtual reality simulation overview.
Source: The American Journal of Medicine, 137(5), 454–458.
https://doi.org/10.1016/j.amjmed.2024.02.002

Figure 2: Virtual reality simulation screenshot.
Source: The American Journal of Medicine, 137(5), 454–458.
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Results

Key Findings:

Surgical Precision: Significant improvements in surgical accuracy were observed among participants (Garber et al., 2024).

Diagnostic Skills: Enhanced ability to diagnose complex conditions through repeated VR simulations.

Student Feedback: Overwhelmingly positive responses, with students reporting increased confidence and readiness for clinical practice.

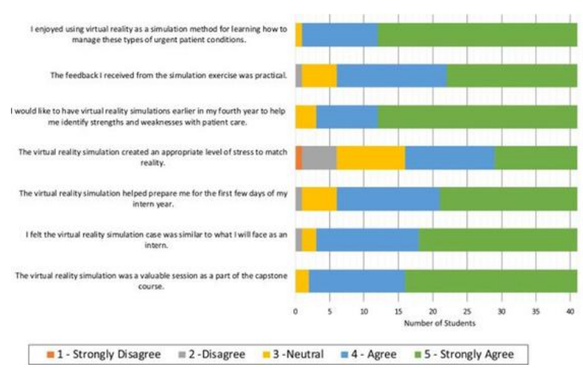

Figure 3: Medical student virtual reality simulation survey results (n = 41).
Source: The American Journal of Medicine, 137(5), 454–458.
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Discussion

Impact of VR: VR indeed attends to many limitations of traditional medical training. Many medical curricula across institutions could easily integrate a scalable, cost-effective solution into their programs (Hayre et al., 2020).

Cost-Effectiveness: Compared to conventional high-fidelity simulations, VR can be said to be very cost-effective. It is a more approachable training tool that requires much fewer resources while delivering the same or, made plain, equal, if not better, results.

Future Directions

AR Integration: Building on the significant progress made in VR, the next step will be the integration of Augmented Reality (AR), which will further enhance training experiences. AR emphasizes real-time data and guides the process, adding an extra layer to make procedures more comprehensible.

Ongoing Research: Continued research and investment will focus on maximizing VR and AR's benefits in medical education. Future studies should explore long-term outcomes and the scalability of these technologies across diverse learning environments and settings.

Conclusion

The increased skill demonstrated in this study will hit the potential of VR as core to medical education and lead us to develop even better and more efficient training programs for integrating VR and AR.

References

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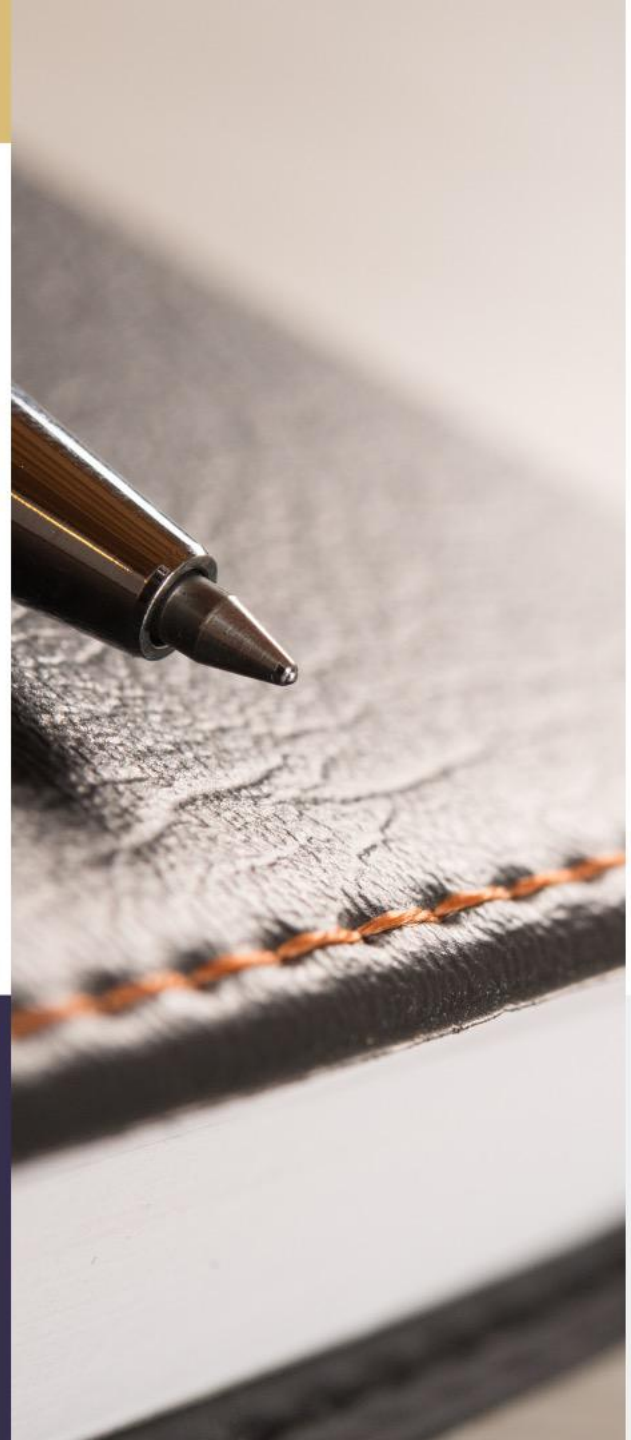

Agenda

1. Introduction
2. Revolutionizing Medical Training with Virtual Reality
3. Assessing VR's Impact on Medical Training
4. Innovative Methods in VR Training
5. Significant Outcomes of VR Training
6. Broad Implications of VR in Medical Training
7. The Future of VR in Medical Education
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Introduction

Overview of VR in Medical Education:

Virtual Reality (VR) is transforming medical training.

Purpose of the Study:

To evaluate the effectiveness of VR in enhancing surgical skills, diagnostics, and patient management.

Significance:

Explore the implications for medical education and the potential future of VR integration in curricula.

Revolutionizing Medical Training with Virtual Reality

- **VR revolutionizes medical training by** bringing immersive, hands-on experiences to medical education (Garber et al., 2024).
- **Enhances surgical and diagnostic skills:** VR provides a safe space to practice complex procedures.
- **Improves patient management training:** Offers realistic scenarios for better patient care strategies.

Assessing VR's Impact on Medical Training

- **Evaluating effectiveness:** Focusing on key areas—surgery, diagnostics, and patient interaction.
- **Measurable outcomes:** Quantitative data demonstrates significant skill improvements.

Innovative Methods in VR Training

- **Transforming simulations:** Converting traditional high-fidelity cases into immersive VR experiences.
- **360-degree video immersion:** Fully engaging students in a realistic environment.
- **Student feedback:** Valuable insights from those closest to entering the field (Chinnock, 1994).

Significant Outcomes of VR Training

- **Enhanced precision and diagnostics:** Noticeable improvements in surgical and diagnostic skills (Garber et al., 2024).
- **Positive feedback:** Students find VR invaluable in preparing for real-world practice.
- **Increased readiness:** VR boosts confidence and competence for future medical professionals.

Broad Implications of VR in Medical Training

- **Bridging gaps:** Addressing limitations in traditional training methods.
- **Cost-effective and scalable:** Making high-quality training accessible to more institutions (Hayre et al., 2020).
- **Future integration:** Potential for broader adoption in medical curricula.

The Future of VR in Medical Education

A comprehensive tool:

VR is more than just an add-on—it is a game-changer

Ongoing research needed:

Continued exploration and investment are crucial.

AR integration:

The next step in evolving medical training

(Garber et al., 2024).

Conclusion

Key Points:

- VR's impact on medical training
- The benefits of VR in enhancing skills
- The future potential of VR and AR in medical education

Call to Action:

- Further research and integration of VR into medical curricula.

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- **Chinnock, C. (1994).** Virtual reality in surgery and medicine. *PubMed*, 13(18), 1–48. https://pubmed.ncbi.nlm.nih.gov/10172193

- **Garber, A. M., Meliagros, P., Diener-Brazelle, J., & Dow, A. (2024b).** Using Virtual Reality to Teach Medical Students Cross-Coverage Skills. *The American Journal of Medicine*, 137(5), 454–458. https://doi.org/10.1016/j.amjmed.2024.02.002

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Q&A

Thank You

For further inquiries and comments,